Green Synthesis of Activated Carbon from Agricultural Biomass Waste

Utilizing Tea Waste and Banana Plant Extract from North-East India for Sustainable Carbon Material Production

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Region of Focus:	North-East India (Assam)
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Feedstocks: Tea processing waste | Banana plant residues

Applications: Activated carbon for energy storage & filtration

Key Innovation: Bio-chemical activation replacing corrosive KOH

Abstract

India generates approximately 500–550 million tonnes of agricultural biomass residues annually, a significant portion of which is either burned or discarded without productive use. This report proposes a green synthesis pathway for producing high-surface-area activated carbon by repurposing two co-located agricultural waste streams in North-East India: tea processing residues and banana plant biomass. Banana plant extract, which is naturally rich in potassium-based compounds, replaces the corrosive chemical activating agents commonly used in conventional production methods. The proposed approach reduces environmental hazards, lowers production costs, and supports regional circular economy development. Resulting activated carbon materials are suitable for applications in water purification, air filtration, and electrochemical energy storage (supercapacitors).

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1. Introduction

This document serves as the consolidated technical and strategic report for *Project atharvam*, submitted as part of the Startup Sprint 2026 competition. It integrates the original research proposal with a detailed account of the sprint execution journey, the technical architecture of the proposed synthesis process, and the operational milestones achieved between February 4 and March 9, 2026.

1.1. Energy Storage and the Role of Supercapacitors

The global transition toward renewable energy systems has intensified demand for high-performance energy storage technologies. Electric Double-Layer Capacitors (EDLCs)—commonly referred to as supercapacitors—are emerging as a critical complement to conventional battery systems. These devices store energy through electrostatic charge accumulation at the electrode–electrolyte interface, enabling rapid charge-discharge cycles, long operational lifetimes, and high power densities. Their performance is strongly governed by the structural and chemical properties of the electrode material.

1.2. Activated Carbon as the Dominant Electrode Material

Activated carbon is the dominant electrode material in commercial EDLCs owing to its high specific surface area, tunable pore structure, and low cost relative to alternative materials. However, most industrial activated carbon is produced from fossil-derived precursors—such as coal and petroleum coke—using chemical activating agents including potassium hydroxide (KOH), phosphoric acid (H_3PO_4), and sulfuric acid (H_2SO_4). This reliance on non-renewable feedstocks and hazardous reagents represents both an environmental liability and a supply chain vulnerability.

1.3. Circular Economy Approach in Materials Science

The identification of agricultural residues as alternative carbon precursors has gained significant attention in recent literature. Biomass-derived carbons from lignocellulosic feedstocks possess inherently hierarchical pore structures—a direct result of the biological architecture of plant cell walls—that are well-suited to supercapacitor electrode applications.

The overarching objective of Project atharvam is to validate the transition from a **Linear Material Failure**—characterized by import-heavy, fossil-derived carbon—to a **Circular**

Resource Model anchored in locally sourced agricultural waste.

2. Problem Analysis: The Linear Material Failure

2.1. Import Dependency and Economic Vulnerability

India's energy storage sector is presently constrained by a structural supply chain paradox. The transition to Electric Vehicles (EVs) and renewable-heavy power grids requires EDLCs that incorporate high-surface-area activated carbon. Yet in 2024, India's import of activated carbon exceeded $98.2 million. The vast majority of this material is derived from fossil-fuel-intensive manufacturing processes in foreign markets, creating two compounding risks:

- **Economic Vulnerability:** Exposure to global commodity price fluctuations, import duties, and logistics delays, all of which inflate the cost of a material positioned as foundational to clean energy infrastructure.
- **Sustainability Gap:** A device marketed as a clean energy technology (the EV supercapacitor) is underpinned by a carbon-heavy manufacturing process—a contradiction that undermines India's stated climate commitments.

A targeted domestic market share of 15% (≈$14.7 million) represents a viable initial commercial objective for a biomass-derived activated carbon startup.

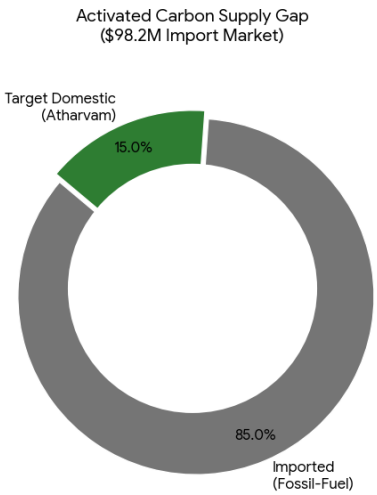

Figure 1. *Activated carbon supply gap in India's $98.2M import market. A dominant 85% of supply is currently met through fossil-fuel-derived imports, while the target domestic share for the atharvam model (Atharvam) is 15%, representing approximately $14.7M in near-term addressable revenue.*

2.2. The Methane Crisis in Agricultural Waste Management

India's agricultural sector generates in excess of 500 million tonnes of biomass residues annually. In North-East India, tea processing waste and banana plant residues routinely undergo unmanaged open-air decomposition or anaerobic breakdown according to the general reaction:

$$\text{Biomass} \xrightarrow{\text{Anaerobic}} CH_4 + CO_2 \tag{1}$$

Methane (CH_4) carries a global warming potential approximately 84 times that of CO_2 over a 20-year horizon. The majority of this lignocellulosic biomass remains poorly utilized—despite being naturally rich in the hierarchical carbon structures required for high-performance electrode applications.

2.3. Environmental Cost of Conventional Activated Carbon Production

Commercial activated carbon production introduces several systemic drawbacks:

- **Fossil Feedstock Dependence:** Coal and petroleum coke are non-renewable resources whose extraction contributes to upstream environmental degradation and

greenhouse gas emissions.

- **Chemical and Energy Intensity:** Activation with KOH requires temperatures in excess of 700 °C. The simplified governing reaction is:

$$6\,\mathrm{KOH} + \mathrm{C} \;\rightarrow\; 2\,\mathrm{K} + 3\,\mathrm{H_2} + 2\,\mathrm{K_2CO_3} \tag{2}$$

  While effective at generating microporous structures, this process introduces significant safety and disposal challenges.

- **Process Emissions:** Activation and washing stages generate chemical effluents requiring costly treatment prior to discharge.

3. Opportunity Assessment

3.1. India's Biomass Waste Resource Potential

India generates approximately 500–550 million tonnes of agricultural biomass residues annually, with an estimated 20–25% burned or discarded without productive use. Conversion of these residues into industrial materials is emerging as a strategic pathway for simultaneous resource efficiency gains and emission reductions within circular economy frameworks.

3.2. North-East India as a Concentrated Biomass Resource Base

3.2.1. Tea Production

Assam alone produces over 650 million kilograms of tea annually—more than 50% of India's total national output. During tea processing, approximately 30–40% of material becomes waste or low-grade residue, constituting a steady, year-round supply of lignocellulosic carbon precursor.

3.2.2. Banana Production

India is the world's largest producer of bananas, contributing approximately 26–28% of global output. Considerable quantities of banana plant residues—particularly stems and peels—remain unused at the farm level and frequently undergo unmanaged decomposition across the North-East.

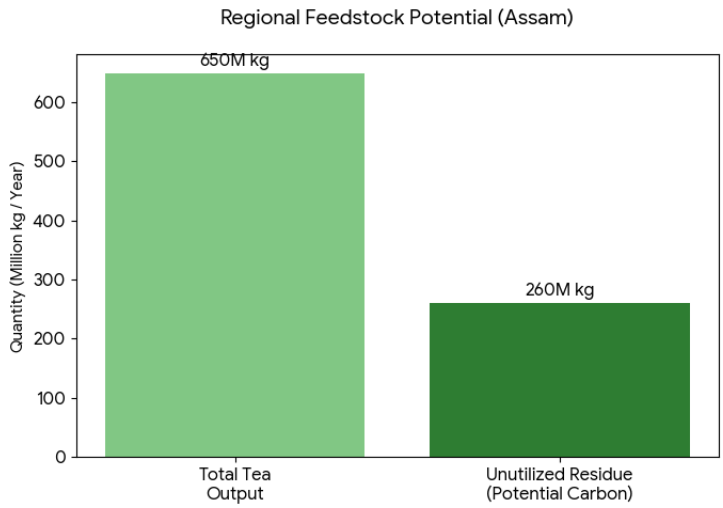

Figure 2. *Regional feedstock potential in Assam (million kg/year). Of the 650 million kg of tea produced annually, approximately 260 million kg (40%) constitutes unutilized residue—representing a large-scale, temporally stable supply of activated carbon precursor material that currently goes to waste.*

3.3. Expanding Demand for Activated Carbon

Key application domains include water purification, air filtration, environmental remediation, and electrochemical energy storage. Water treatment alone accounts for approximately 40–50% of global activated carbon demand. With tightening environmental regulations and the rapid expansion of clean energy technologies, demand is projected to grow at a compound annual rate of 8–11% through 2030.

4. Proposed Solution: Green Synthesis of Activated Carbon

4.1. Core Concept

The proposed solution leverages the co-located biomass streams to produce high-surface-area activated carbon through an entirely green synthesis pathway:

- **Tea waste** serves as the primary carbon precursor, providing a lignocellulosic matrix with inherent porosity and high carbon content.
- **Banana plant extract** (potassium-rich ash leachate) functions as the natural activat-

ing agent, replacing synthetic KOH with naturally occurring potassium carbonate (K_2CO_3).

4.2. Bio-Chemical Activation Mechanism

The proposed process replicates conventional KOH activation using potassium carbonate derived from banana plant ash:

$$\text{Banana Plant Biomass} \xrightarrow{\Delta} K_2CO_3 + \text{other mineral oxides} \tag{3}$$

When the K_2CO_3-rich extract is impregnated into tea waste and subjected to thermal carbonization, it facilitates pore development through a mechanism analogous to conventional chemical activation—without requiring the synthesis, transport, or storage of corrosive reagents.

4.3. Process Overview

1. **Feedstock Collection.** Tea processing waste—including tea dust and rejected leaves—is collected from estates and processing facilities in Assam.

2. **Cleaning and Drying.** Tea waste is washed to remove impurities and dried to a constant weight, ensuring uniform moisture conditions prior to carbonization.

3. **Preparation of Banana Extract.** Banana plant residues are subjected to controlled combustion to produce potassium-rich mineral ash, which is then leached in deionized water and filtered to yield the activating extract.

4. **Impregnation.** Dried tea powder is immersed in the banana extract, enabling potassium compounds to infiltrate the biomass structure at the cellular level.

5. **Thermal Carbonization.** The impregnated biomass is heated at 600–800 °C in an inert atmosphere. Potassium compounds catalyze pore formation, yielding a highly porous carbon structure with elevated specific surface area.

6. **Washing and Refining.** The material is washed with deionized water to remove residual mineral salts, producing a clean, porous activated carbon framework.

4.4. Advantages of the Proposed Method

Environmental Sustainability

The process eliminates corrosive chemical activating agents, replacing them with biomass-derived potassium compounds. This reduces chemical hazards, eliminates the need for specialist wastewater treatment, and lowers the environmental risk profile of the manufacturing operation.

Waste-to-Resource Conversion

Agricultural residues that would otherwise undergo unmanaged anaerobic decomposition are transformed into a high-value industrial material, directly reducing waste-related methane and CO_2 emissions at the source.

Economic Feasibility

Raw materials are inexpensive, widely available, and regionally sourced. Eliminating synthetic activating agents and using low-cost agricultural waste as the carbon precursor substantially reduces input costs relative to conventional production.

5. Sprint Journey: Chronological Execution

The sprint was structured into four distinct phases, navigating through academic constraints and technical validation requirements.

5.1. Phase 1: Discovery and Resource Mapping (February 4–10)

Objective: Confirm that the scale of rejected tea material in Assam justifies a commercially viable feedstock supply chain.

Key Activities:

- Mapping of tea estate concentration across Assam and corresponding waste output volumes per processing facility.
- Literature survey on lignocellulosic carbon precursors and their suitability for activated carbon production.

- Initial scoping of banana plant residue availability and seasonal production cycles.

Outcome: Confirmed that 30–40% of processed tea material in Assam is rejected as low-grade dust—an estimated 260 million kilograms per year available as carbon precursor (see Figure 2).

5.2. Phase 2: Technical Protocol Design (February 11–20)

Objective: Identify a compatible and regionally available activating agent for tea-derived activated carbon.

Key Activities:

- Systematic review of biomass ashes and their potassium content profiles.
- Evaluation of banana plant residues as a potassium carbonate source.
- Review of Indian Patent No. 377645, describing the use of banana plant ash extract (*khar*) as a natural activating agent for tea-derived activated carbon—directly validating the core innovation.
- Preliminary design of impregnation ratios and carbonization temperature profiles.

Outcome: A patent-validated, technically grounded activation protocol was established, defining the five-step process flow described in Section 4.

5.3. Phase 3: Strategic Hiatus and Passive Review (February 21 – March 6)

Active prototyping was paused for 14 days due to mid-semester academic examination commitments. This period yielded a critical operational insight: the temporal asynchrony between tea waste (year-round availability) and banana plant residues (seasonally concentrated supply).

Activities during the hiatus:

- Passive review of literature on hierarchical carbon morphology.
- Refinement of theoretical impregnation ratios.
- Development of a **Seasonal Stockpiling Strategy**: banana plant ash is produced and stored during peak harvest periods, enabling year-round activation of the con-

tinuously available tea waste feedstock.

Key Insight ("The Midsem Insight"): The most operationally complex challenge is not the chemistry—which is well-supported by literature—but the management of seasonal feedstock availability. This insight materially refined the supply chain design.

5.4. Phase 4: High-Velocity Finalization (March 7–9)

Post-examination, the sprint resumed with focused documentation of the operational model and finalization of this journey report.

Key Activities:

- Documentation of the full decentralized collection and centralized refinery architecture.
- Preparation of the synthesis process metrics framework.
- Drafting and finalization of this consolidated sprint report.

6. Technical Methodology and Laboratory Execution Plan

6.1. Synthesis Process Metrics Framework

Six key performance indicators were defined to govern quality and efficiency at each stage of the synthesis pipeline. Figure 3 presents these metrics in their relational framework, illustrating how each indicator maps onto a distinct stage of production.

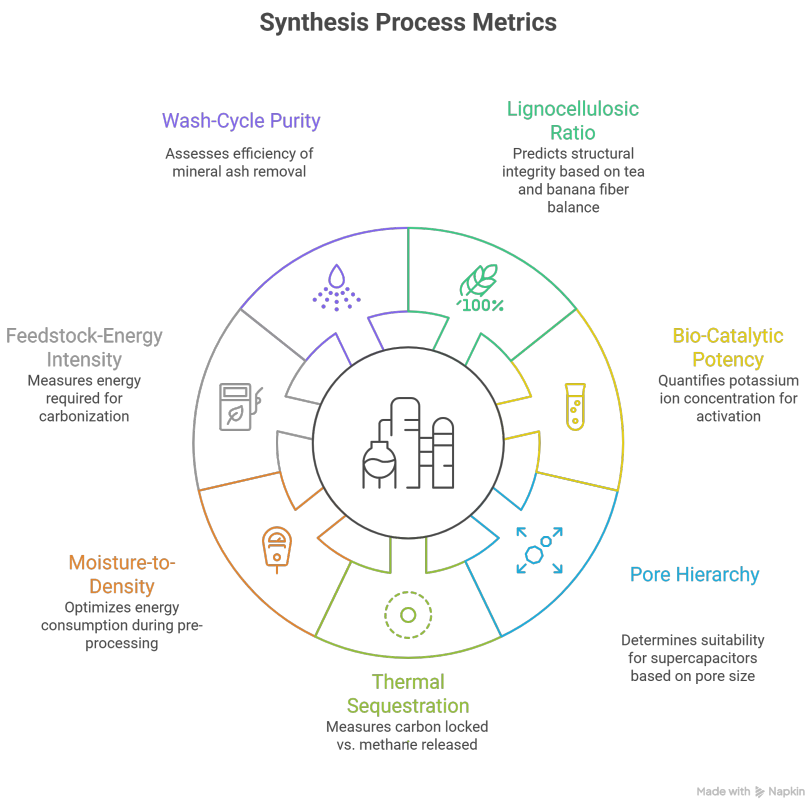

Figure 3. *Synthesis process quality metrics framework for Project atharvam. The six indicators—Lignocellulosic Ratio, Bio-Catalytic Potency, Pore Hierarchy, Thermal Sequestration, Moisture-to-Density, and Wash-Cycle Purity—collectively govern the structural integrity, activation efficiency, and environmental performance of the activated carbon product at each stage of the production process.*

Table 1. *Synthesis Process Quality Metrics — Definitions and Targets*

Metric	Description	Target
Lignocellulosic Ratio	Ratio of tea to banana fibre in the impregnation mix; predicts structural integrity of the final carbon matrix.	Optimised per batch
Bio-Catalytic Potency	Quantifies potassium ion concentration in the banana ash extract prior to impregnation, governing activation efficacy.	$\geq$0.5 mol/L K^+
Pore Hierarchy Index	Characterises the distribution of micropores, mesopores, and macropores; determines suitability for supercapacitor and filtration applications.	BET $\geq$1500 m^2/g
Moisture-to-Density Ratio	Optimises pre-processing drying energy consumption by relating feedstock moisture content to bulk density.	$\leq$5% moisture
Feedstock-Energy Intensity	Measures thermal energy consumed per kilogram of activated carbon produced during carbonization.	Minimised
Wash-Cycle Purity	Assesses efficiency of residual mineral ash removal through conductivity measurement of wash effluent.	$\leq$50 μS/cm

6.2. Utilizing IIT Guwahati Laboratory Infrastructure

Small-scale validation can be conducted through the Technology Incubation Centre (TIC) at IIT Guwahati. Available infrastructure maps directly onto each synthesis stage:

- **Feedstock Stabilization:** Laboratory drying ovens and analytical balances for moisture removal and constant-weight determination.

- **Pre-processing and Size Reduction:** Grinding equipment and sieve systems for uniform biomass particle sizing.
- **Activator Preparation:** Controlled combustion chambers for banana plant ash production, followed by aqueous leaching and vacuum filtration.
- **Carbonization:** High-temperature tube furnaces operating at 600–800 °C under inert nitrogen atmosphere.
- **Characterization:** BET surface area analysis, scanning electron microscopy (SEM), and X-ray diffraction (XRD) accessible through shared university instrumentation facilities.

7. Logistical Model and Regional Integration

7.1. The North-East Bio-Refinery Network

During the sprint, a preliminary architecture was developed for a decentralized **North-East Bio-Refinery Network** connecting agricultural waste streams with carbon material production. The model separates early-stage processing from final carbon synthesis to address the high volume-to-weight ratio of raw biomass:

- **Satellite Pre-processing Units:** Small-scale drying and grinding facilities near tea estate clusters in Assam, reducing biomass volume prior to transport.
- **Central Refinery:** A single centralized facility for high-temperature carbonization, post-processing, quality control, and packaging, enabling economies of scale.

7.2. Competitive Positioning

Table 2. *Comparative Analysis: Conventional vs. atharvam Production Model*

Feature	Conventional (Imported)	atharvam Model
Carbon Precursor	Coal / Petroleum coke	Agricultural waste (tea dust)
Activating Agent	Synthetic KOH	Banana plant ash extract
Environmental Impact	High CO_2; chemical effluent	Negative methane; no hazardous by-product
Logistics Cost	High (import duties)	Low (regional sourcing)
Resource Model	Linear	Circular
Supply Chain Risk	High (import dependency)	Low (local biomass)
Feedstock Cost	Commodity-priced	Near-zero (waste material)

7.3. Economic Feasibility

- **Near-Zero Precursor Cost:** Tea dust and banana plant waste are available at negligible cost as agricultural disposal liabilities.
- **Reduced Transportation Costs:** Decentralized pre-processing eliminates the need to transport bulky, high-moisture raw biomass over long distances.
- **Elimination of Chemical Input Costs:** Replacing KOH with banana ash extract removes a significant recurring reagent cost and associated safety compliance overhead.
- **Additional Rural Value Streams:** Converting residues into industrial feedstock creates supplementary income for plantation operators and smallholder farmers.

7.4. Regional Development Impact

- **Employment Generation:** Satellite units and the central refinery create technical and operational employment in a region with limited industrial diversification.
- **Waste Utilization:** Large volumes of rejected tea dust are redirected from disposal into a revenue-generating industrial input.
- **Localized Industrial Development:** Linking the agricultural economy of Assam

with the clean energy materials sector supports the emergence of a sustainable bio-materials ecosystem in the North-East.

8. Budget Framework

A preliminary budget for the laboratory validation phase has been developed across four expenditure categories, with the allocation designed to concentrate resources in the highest-impact technical activities. Figure 4 presents the full breakdown.

Category	Allocation	Amount	Key Expenditure Items
Essential R&D & Lab	55%	₹ 4,12,500	Small-scale muffle furnace, chemical glassware, and outsourcing advanced testing (CV/GCD) to university labs.
Feedstock & Logistics	15%	₹ 1,12,500	Manual collection from 2 local tea estates; basic solar-drying setup for biomass stabilization.
Ops & Intern Stipends	20%	₹ 1,50,000	2 Part-time research interns; founder-led execution.
Provisional IP & Misc	10%	₹ 75,000	Provisional patent filing and basic material characterization fees.

Figure 4. *Preliminary validation phase budget allocation across four expenditure categories. The total seed requirement is `7,50,000. A majority allocation of 55% is directed toward Essential R&D and Laboratory equipment, reflecting the experiment-intensive nature of the early-stage synthesis validation programme. The remaining budget covers feedstock logistics, intern stipends, and provisional intellectual property filing.*

The four budget categories and their rationale are summarized below:

- **Essential R&D & Lab (55%, `4,12,500):** Small-scale muffle furnace, chemical glassware, and outsourced advanced characterization (cyclic voltammetry and galvanostatic charge-discharge) to university laboratories.

- **Feedstock & Logistics (15%, `1,12,500):** Manual collection from two local tea estates; basic solar-drying setup for biomass stabilization.
- **Operations & Intern Stipends (20%, `1,50,000):** Two part-time research interns supporting founder-led experimental execution.
- **Provisional IP & Misc (10%, `75,000):** Provisional patent filing and basic material characterization fees.

9. Future Roadmap

Following the formal conclusion of this sprint on March 9, 2026, the following milestones define the project trajectory:

Table 3. *Post-Sprint Development Milestones*

Timeline	Milestone
Q2 2026	Laboratory-scale synthesis optimization targeting specific surface area (SSA) of $\geq$1500 m^2/g via BET analysis. Systematic variation of carbonization temperature, impregnation ratio, and dwell time.
Q3 2026	Pilot feedstock partnership established with a regional tea estate for on-site waste collection. First kilogram-scale synthesis run conducted.
Q4 2026	Prototype activated carbon electrodes fabricated and tested in a 10 F supercapacitor cell; electrochemical performance characterization via cyclic voltammetry and galvanostatic charge-discharge.
Q1 2027	Provisional patent filing on activation methodology. Engagement with industrial partners in the EV and water treatment sectors.

10. Conclusion

Project atharvam has progressed from a theoretical concept grounded in circular economy principles into a validated operational roadmap with a technically sound founda-

tion, a commercially compelling market positioning, and a clearly defined development trajectory.

The confluence of three structural conditions—abundant regional biomass, a demonstrable import dependency in the activated carbon market (Figure 1), and the availability of natural activating compounds within the same feedstock region (Figure 2)—creates an exceptionally favorable environment for the proposed green synthesis framework. The synthesis process metrics framework (Figure 3) provides a rigorous, measurable quality control structure to guide laboratory validation through to pilot-scale deployment. The preliminary budget (Figure 4) demonstrates that the validation programme can be executed within a lean, well-prioritized financial envelope.

The sprint has demonstrated that the solution to India's $98.2 million activated carbon import dependency is currently being discarded as refuse on the floors of Assam's tea estates. By linking the tea industry to the emerging EV and energy storage sector through a circular, chemistry-driven supply chain, Project atharvam is positioned to transform a significant economic liability into a sustainable, regionally anchored industrial capability.

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Project atharvam — North-East India Circular Economy Initiative.